

Jasper Sands

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Quantum algorithms, error correction, and machine learning for qubit control. MS at Columbia, quantum computing track — thesis at Diraq and UNSW Sydney on automating silicon spin qubit measurement.

EDUCATION

Columbia University, MS in Computer Science — Quantum Computing Track · GPA 3.91/4.0 Sep 2025 – May 2027

Thesis: automating silicon spin qubit measurement with machine learning — Diraq and UNSW Sydney, Sep 2026 – May 2027, co-supervised by Dr. Nard Dumoulin Stuyck and Prof. Henry Yuen.

Coursework: Quantum Error Correction · Quantum Engineering · Quantum Simulation & Computing Lab. **TA:** Intro to Quantum Computing.

Washington University in St. Louis, BS in Computer Science and Economics · GPA 3.81/4.0

Aug 2021 – May 2025

Coursework: Machine Learning · Analysis of Algorithms · Computer Security. **TA:** Malware Analysis; Parallel Programming.

SELECTED PROJECTS

WhatTheDuck — quantum amplitude estimation for financial Value at Risk · *CUDA-Q, C++* iQuHACK 2026 — Overall Winner

- Wrote the GPU-accelerated state preparation, threshold oracle, and bisection search behind Iterative Quantum Amplitude Estimation; measured query scaling matched the $O(1/\epsilon)$ bound against an $O(1/\epsilon^2)$ quasi-Monte Carlo baseline.

WASM QEC Simulator — surface-code decoding in the browser · *Rust, WebAssembly*

qcompiler.jaspersands.com

- Symplectic stabilizer simulator for rotated and XXXX surface codes, with exact MWPM and Union-Find decoding running client-side; finite-size scaling over $d = 3, 5, 7$ at 20k shots/point puts the phenomenological threshold at 3.2%.

Noise-Aware Adiabatic Simulation — transverse-field Ising model · *Qiskit*

IBM Heron hardware

Digitized adiabatic ground-state preparation on Heron's native fractional RZZ gates, roughly halving logical-to-physical translation error, with palindromic Trotter steps and a schedule flattened where the gap closes.

Q-Search — multi-agent quantum algorithm discovery engine · *Python, Gemini*

qsearch.jaspersands.com

Agents propose problems, design circuits, verify them by stabilizer and oracle simulation, and fit complexity scaling across 795 runs.

LLM-Based Lossless Text Compression · *PyTorch, CUDA*

Systems optimization

Arithmetic coding on autoregressive LLM predictions: 5–16× average and 39× best-case compression, beating zstd and gzip; static KV caching and CUDA graphs for throughput.

RESEARCH

Cidon Systems Research Lab, Columbia University · *Graduate Researcher*

Sep 2025 – Jun 2026

- Built phishing and spam detection pipelines with Dr. Asaf Cidon and Barracuda Networks, combining language-model embeddings with production email telemetry and tuning for precision against a fixed false-positive budget.

Complex Resilient Intelligent Systems Lab, Columbia University · *Graduate Researcher*

May – Dec 2025

- Reduced hallucination in generative protein modeling under Dr. Venkat Venkatasubramanian by grounding outputs in molecular graph data.

Foster Care Policy Database, Washington University in St. Louis · *Undergraduate Researcher*

Sep 2024 – Jun 2025

- Fine-tuned LLaMA-3 with Unsloth and Hugging Face for QA over 50k foster-care policy documents, with human-in-the-loop feedback.

Also: Desdr Open Insurance Toolkit, Columbia (Dr. Eugene Wu, Sep–Dec 2025) — SMS survey data into actuarial and climate-risk features for agricultural loss estimation. **Computer Vision & AI Lab**, WashU (Dr. Umar Iqbal, Sep 2024–Jun 2025) — distributed pipelines parsing billions of forum posts.

EXPERIENCE

Smack Technologies · *Software Engineer Intern*

May – Aug 2026

- Wrote physics simulations and decision models for the core technology pipelines, plus benchmarks validating the learned policies.

Highnote · *Cybersecurity Engineer Intern*

May – Jul 2024

- Designed and deployed the company-wide SIEM for a card-issuing fintech, unifying AWS, GCP, and Datadog telemetry into one Elasticsearch cluster ingesting over 1 TB/day, and wrote 100+ rule-based and ML anomaly detections.

Mindtrip · *Software Engineer Intern*

May – Jul 2023

- Built internal admin and secure-query tooling in JavaScript and Rails via Retool, saving roughly 250 engineering hours a month.

Earlier: Bugcrowd, Associate Application Security Engineer (May–Jul 2020) — vulnerability triage and client disclosure work.

HONORS & LEADERSHIP

- Overall Winner** and **NVIDIA Ecosystem Award**, MIT iQuHACK 2026 · Qualified, NYUAD Quantum Hackathon for Social Good 2026 · **Audience Favorite**, Hamoniqs Quantum Design Hackathon 2026 · **Runner-Up**, Qualcomm Snapdragon Multiverse Hackathon 2026
- Graduate Lead**, Columbia Quantum Algorithms Reading Group (Sep 2025–Jun 2026) — weekly sessions on Hamiltonian simulation and LCU.

SKILLS

Quantum: Qiskit, Cirq, PennyLane, CUDA-Q · Hamiltonian simulation, phase estimation, amplitude amplification and estimation, QSVT and block encoding, LCU · surface codes, MWPM and Union-Find decoding · VQE, QAOA · noise modeling and transpilation

Languages: Python, C++, Rust, C, Go, Java, TypeScript, MATLAB

ML & systems: PyTorch, TensorFlow, scikit-learn, Hugging Face, CUDA · Linux, Git, Docker, WebAssembly, AWS, GCP, Elasticsearch, LaTeX

Mathematics: Linear algebra, spectral graph theory, tensor networks, probability, optimization, quantum complexity (BQP, QMA)